

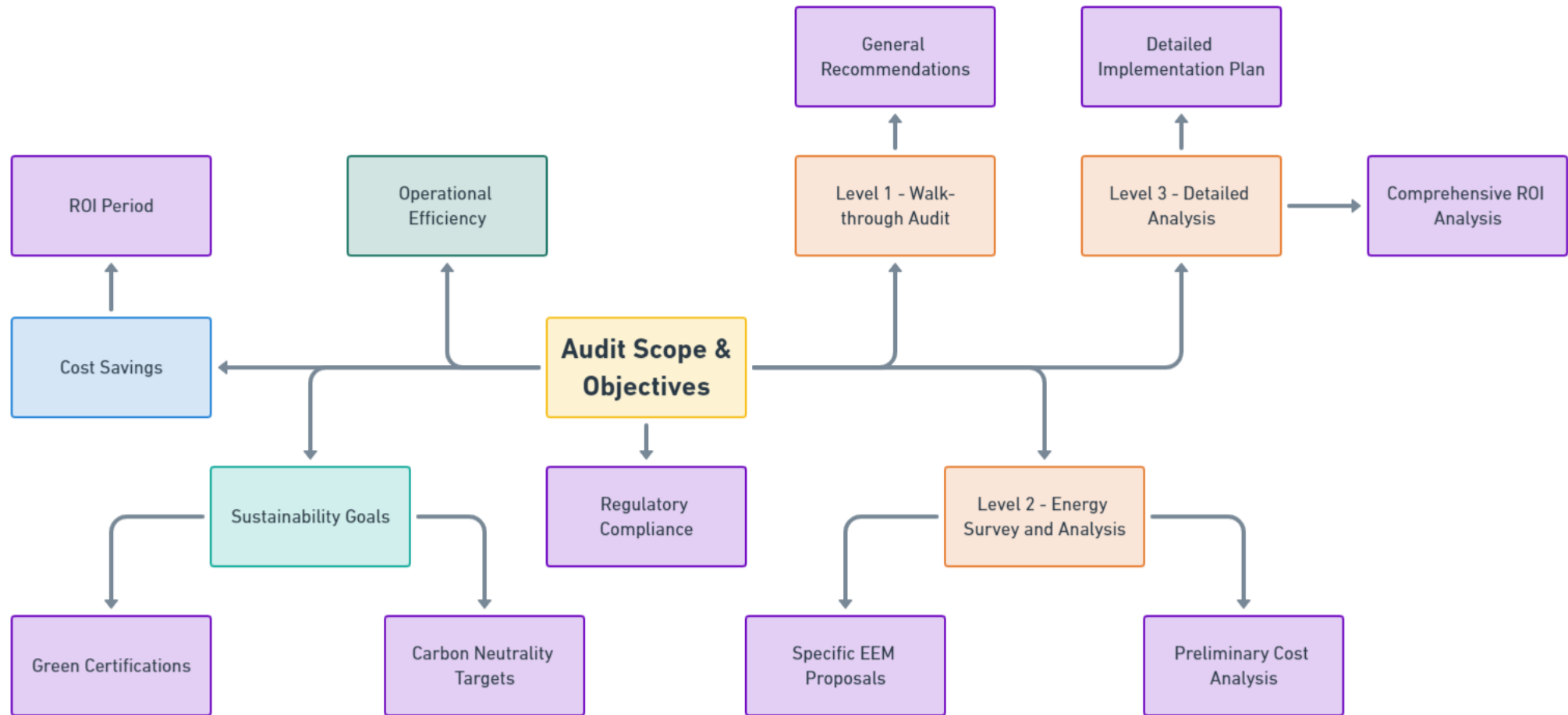
# Best Practices for Energy Audits Using ASHRAE 211-2018 and RETScreen

Session F3: Energy Auditing

Javier Ojeda, P.Eng, CEM, REP, PMP, MBA

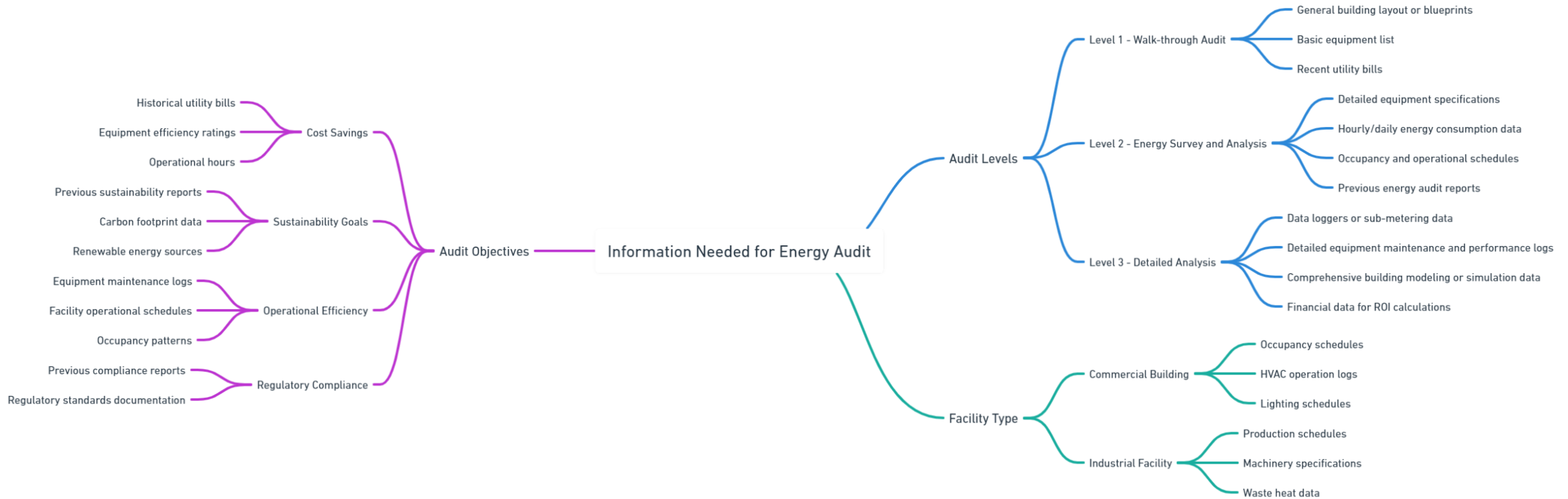
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# Planning the Audit



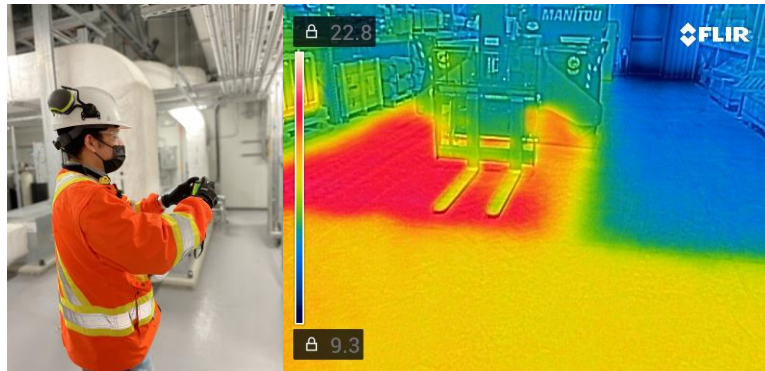
Made with Whimsical

# Planning the Audit



Made with Whimsical

# Planning the Audit



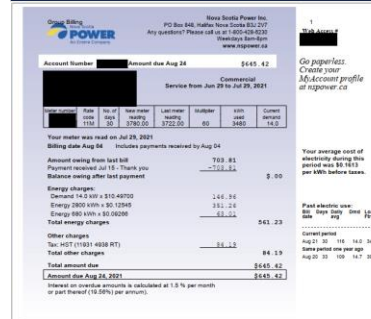
TOOLS FOR THE  
ENERGY AUDIT



# Conducting the Audit

- Site Visits
  - Stick to your plan... unless reality hits you in the face!

## Documentation



## SAFETY toolbox meeting



## Preliminary Walkthrough

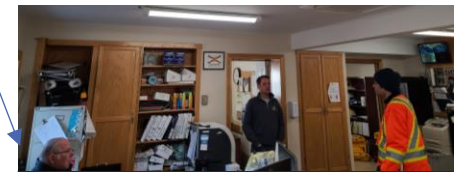
## Equipment Checks



## Data Collection



## Interviews



# Conducting the Audit

RETScreen Expert - Digby - 5 Bay Garage - NSDPW - COMBINED.rctx

File Location Facility Energy Cost Emission Finance Risk Report Custom

Benchmark Feasibility Performance All Level 1 Level 2 Hide benchmark Benchmark database... Show notes Show image Default image Select image... Adjust image... Keep aspect ratio Help eLearning Workflow... Colour coding...

Step 1 - Analysis type Step 2 - Benchmark Options Help

Subscriber: Hysovent Limited - Professional

RETScreen - Facility

Facility information

Facility type: Commercial/Institutional  
Type: Other  
Description: Commercial/Institutional - Maintenance Shop

Prepared for: Nova Scotia Department of Public Works  
Prepared by: Hysovent Limited

Facility name: Digby - 5 Bay Garage  
Address: 200 Robinson Weir Rd  
City/Municipality: Digby  
Province/State: NS  
Country: Canada

Benchmark - Commercial/Institutional - Other

Facility size: 589 m<sup>2</sup>

Fuel consumption

Energy use intensity: GJ/m<sup>2</sup>  
Reference unit: m<sup>2</sup>

Benchmark: 0.82 GJ/m<sup>2</sup>  
Minimum (Typical): 0.62 GJ/m<sup>2</sup>  
Maximum (Typical): 2.16 GJ/m<sup>2</sup>

Base case: 1.44 GJ/m<sup>2</sup>  
Reference year: 2022

Set target: Target  
Year: 2030  
Target: -50%  
Proposed case: 0.72 GJ/m<sup>2</sup>

Facility - Plan: Annual

Energy use intensity (GJ/m<sup>2</sup>)

Plan

Base case

Target

Benchmark

Net zero

Let's go with Live Example!

- Data Analysis – RETScreen + Excel + ASHRAE Forms

All Audit Levels - Annual Summary

Summary for Existing Building

user input  
calculated

Imported Energy (E<sub>imp</sub>)

| Energy Type                  | Annual | Units                | Conversion | Mass           | Annual Cost      |
|------------------------------|--------|----------------------|------------|----------------|------------------|
| Electricity                  | 47,071 | kWh                  | 3.4        | 172,235        | \$ 7,873         |
|                              | -      |                      |            |                | \$ -             |
|                              | -      |                      |            |                | \$ -             |
|                              | -      |                      |            |                | \$ -             |
|                              | -      |                      |            |                | \$ -             |
| Oil                          | 17,276 | Barrel (Fuel Oil #2) | 39.3       | 672,441        | \$ 15,211        |
|                              | -      |                      |            |                | \$ -             |
|                              | -      |                      |            |                | \$ -             |
| <b>E<sub>imp</sub> Total</b> |        |                      |            | <b>844,676</b> | <b>\$ 23,084</b> |

On-Site Renewable Energy Production (E<sub>re</sub>)

| Energy Type                 | Annual | Units | Conversion | Mass     | Annual Cost |
|-----------------------------|--------|-------|------------|----------|-------------|
|                             | -      |       |            |          | \$ -        |
|                             | -      |       |            |          | \$ -        |
| <b>E<sub>re</sub> Total</b> |        |       |            | <b>-</b> | <b>\$ -</b> |

Exported Energy (E<sub>exp</sub>)

| Energy Type                  | Annual | Units | Conversion | Mass     | Annual Cost |
|------------------------------|--------|-------|------------|----------|-------------|
|                              | -      |       |            |          | \$ -        |
|                              | -      |       |            |          | \$ -        |
| <b>E<sub>exp</sub> Total</b> |        |       |            | <b>-</b> | <b>\$ -</b> |

Existing Building EUECI

| Building Name                                      | Digby 5 Bay Garage |
|----------------------------------------------------|--------------------|
| Grand Conditional Energy Factor                    | \$99               |
| EUI (Building) (MJ/m <sup>2</sup> /yr)             | 1,438.2            |
| EUI (Site) (MJ/m <sup>2</sup> /yr)                 | 1,438.2            |
| Site ECI (Energy cost index \$/m <sup>2</sup> /yr) | \$ 44.41           |

EUI: Energy Use Intensity

Benchmarking & Target EUECI

| Benchmark Source                              | Value          |
|-----------------------------------------------|----------------|
| Benchmark Source                              |                |
| Benchmark EUI (MJ/m <sup>2</sup> /yr)         |                |
| Target EUI (building) (MJ/m <sup>2</sup> /yr) |                |
| Target EUI (site) (MJ/m <sup>2</sup> /yr)     |                |
| Target ECI (site) (\$/m <sup>2</sup> /yr)     |                |
| Annual Energy Savings to Reach Target         | MJ             |
| Annual Cost Savings to Reach Target           | (\$ available) |

Notes

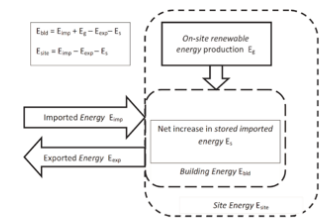
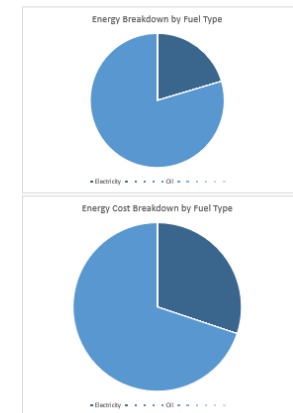


Figure 1 Relationship of site energy (E<sub>site</sub>) and building energy (E<sub>bu</sub>)